🗺️ COMPLETE NAVIGATION & FLOW ARCHITECTURE

🌟 Full Inch-by-Inch User Journey with Every Screen Connection

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🎛️ APP LAUNCH FLOW

1️⃣ COLD START → SPLASH SCREEN

text

User opens app

↓

SPLASH SCREEN loads (2 seconds)

```
    ├── Load Firebase config
    ├── Check local login state
    └── Decide next screen
```

↓

IF userToken exists → AUTO NAVIGATE to SCREEN 6 (Farmer Home)

ELSE → NAVIGATE to SCREEN 2 (Language Selection)

🔄 PHASE 1: ONBOARDING & AUTHENTICATION

2️⃣ LANGUAGE SELECTION SCREEN

text

User sees two large buttons

↓

User taps "தமிழ்" or "English"

↓

Continue button activates

↓

User taps "Continue"

↓

SAVE language preference locally

↓

NAVIGATE to SCREEN 3 (Phone Login)

3️⃣ PHONE NUMBER LOGIN

text

Auto-focus on phone input

↓

User types 10-digit number

↓

"Get OTP" button activates

↓

User taps "Get OTP"

↓

SHOW loading spinner

↓

Firebase sends OTP to phone

↓

NAVIGATE to SCREEN 4 (OTP Verification)

4️⃣ OTP VERIFICATION

text

Auto-focus on first OTP box

↓

User types 6-digit OTP

└── Auto-move between boxes

↓

WHEN all 6 digits entered:

├── Auto-verify OTP with Firebase

├── SHOW verification loading

└── On success → PROCEED to Auto Routing

↓

IF OTP wrong → Show error

↓

Timer counts down from 2:00

↓

AFTER timer ends → Enable "Resend OTP"

5️⃣ AUTO ROUTING (INVISIBLE SCREEN)

text

AFTER OTP verification:

↓

CHECK Firestore: users/{uid}.role

↓

IF role == "owner":

→ NAVIGATE to SCREEN 22 (Owner Dashboard)

ELSE:

→ NAVIGATE to SCREEN 6 (Farmer Home)

↓

SAVE login state locally

🛒 PHASE 2: FARMER E-COMMERCE FLOW

MAIN NAVIGATION BAR (Bottom)

text

[HOME] [PRODUCTS] [ADVISORY] [PROFILE]

6️⃣ FARMER HOME SCREEN

text

Greeting shows: "வணக்கம், [Name]!"

↓

Banner auto-scrolls (offers/alerts)

↓

QUICK ACTIONS GRID:

└── TAP "🛒 Products" → NAVIGATE to SCREEN 7 (Categories)

└── TAP "🌾 My Crops" → NAVIGATE to SCREEN 16 (My Crops)

└── TAP "🔁 Re-Buy" → NAVIGATE to SCREEN 14 (Re-Buy)

└── TAP "📦 My Orders" → NAVIGATE to SCREEN 13 (Order Tracking)

↓

BELL Icon (top-right) → NAVIGATE to Notification Screen

PRODUCT BROWSING PATH

Path A: Via Categories

text

SCREEN 6 (Home) → TAP "Products"

↓

SCREEN 7 (Categories) → TAP "Seeds"

↓

SCREEN 8 (Product List) → TAP Product Card

↓

SCREEN 9 (Product Details) → TAP "Add to Cart"

↓

SCREEN 10 (Cart Screen)

Path B: Direct Search

text

SCREEN 6 (Home) → BOTTOM NAV "Products"

↓

SCREEN 8 (Product List) → USE Search Bar

↓

Tap product → SCREEN 9 → Add to Cart

↓

SCREEN 10 (Cart)

CHECKOUT FLOW

text

SCREEN 10 (Cart) → TAP "Proceed to Checkout"

↓

SCREEN 11 (Checkout)

```
    ├── Review order summary
    ├── Select payment: Cash/Credit
    └── TAP "Place Order"
```

↓

SCREEN 12 (Order Success)

├── TAP "View Orders" → SCREEN 13

└── TAP "Back to Home" → SCREEN 6

ORDER TRACKING FLOW

text

From ANYWHERE → BOTTOM NAV "Profile"

↓

SCREEN 30 (Profile) → TAP "My Orders"

OR

From Home → TAP "My Orders" quick action

↓

SCREEN 13 (Order Tracking)

└── Shows timeline: Reserved → Ready → Picked

🌾 **PHASE 3: ADVISORY SYSTEM FLOW**

CROP MANAGEMENT PATH

text

SCREEN 6 (Home) → TAP "My Crops"

OR

BOTTOM NAV "Advisory" → TAP "My Crops" section

↓

SCREEN 16 (My Crops Dashboard)

├── IF no crops → AUTO NAV to SCREEN 15 (Add Crop)

└── IF has crops → List of crop cards

↓

TAP any crop card → SCREEN 31 (Crop Details)

├── Shows crop age

├── Recommended products section

└── TAP any product → SCREEN 9 (Product Details)

DOSAGE CALCULATOR PATH

text

SCREEN 9 (Product Details) → Scroll to "Dosage" tab

↓

TAP "Calculate for my land"

↓

SCREEN 18 (Dosage Calculator)

```
    ├── Auto-fill crop area from profile
    ├── Calculate required quantity
    └── TAP "Add to Cart" → SCREEN 10 (Cart)
```

ADVISORY ALERTS PATH

text

From ANYWHERE → Notification Bell

OR

BOTTOM NAV "Advisory"

↓

SCREEN 19 (Advisory Messages)

```
    └── List of shop messages
```

↓

TAP any message → SCREEN 20 (Voice Player)

```
    └── Auto-play voice advice
```

PEST ALERTS PATH

text

When owner sends pest alert:

↓

PUSH NOTIFICATION appears

↓

User taps notification

↓

OPEN SCREEN 21 (Pest Alert Details)

```
    ├── Shows affected crop
    ├── Recommended action
    └── TAP "View Products" → SCREEN 8 (Filtered list)
```

👔 PHASE 4: OWNER PANEL FLOW

OWNER DASHBOARD NAVIGATION

text

Owner logs in → AUTO to SCREEN 22

↓

OWNER BOTTOM NAV: [Dashboard] [Orders] [Stock] [Farmers]

ORDER MANAGEMENT PATH

text

SCREEN 22 → TAP "Manage Orders"

OR

BOTTOM NAV "Orders"

↓

SCREEN 23 (Order Management)

└── List of all orders

↓

TAP any order → SCREEN 32 (Order Details)

├── Farmer info

├── Items list

├── BUTTON: "Mark as Ready"

└── BUTTON: "Mark as Picked"

STOCK MANAGEMENT PATH

text

SCREEN 22 → TAP "Manage Products"

OR

BOTTOM NAV "Stock"

↓

SCREEN 24 (Product Stock)

├── List all products with stock

└── TAP "Add Product" → SCREEN 25

↓

TAP any product row → SCREEN 25 (Edit Product)

└── Update price/stock/expiry

FARMER ACCOUNTS PATH

text

SCREEN 22 → TAP "Farmer Accounts"

OR

BOTTOM NAV "Farmers"

↓

SCREEN 26 (Farmer Accounts)

└── List all farmers with credit

↓

TAP any farmer → SCREEN 33 (Farmer Details)

├── Purchase history

├── Credit ledger

└── BUTTON: "Add Credit Entry"

BROADCAST MESSAGES

text

SCREEN 22 → TAP "Send Alert"

↓

SCREEN 28 (Broadcast Alert)

├── Type message/upload voice

├── Select target crops

└── TAP "Send"

👤 PROFILE & SETTINGS FLOW

3 0️⃣ PROFILE SCREEN

text

From ANYWHERE → BOTTOM NAV "Profile"

↓

SCREEN 30 (Profile)

├── Profile info

```
    ├── TAP "My Orders" → SCREEN 13
    ├── TAP "My Crops" → SCREEN 16
    ├── TAP "Language" → SCREEN 2
    ├── TAP "Change Phone" → SCREEN 3 (with logout)
    └── TAP "Logout" → Logout → SCREEN 2
```

🔄 CROSS-FUNCTIONAL FLOWS

RE-BUY FLOW

text

```
SCREEN 6 (Home) → TAP "Re-Buy" quick action
↓
SCREEN 14 (Re-Buy List)
    └── Previous purchases
↓
TAP "Buy Again" on any product
↓
DIRECT to SCREEN 10 (Cart) with item added
```

NOTIFICATION SYSTEM

text

```
When notification arrives:
↓
1. PUSH notification shows
2. App badge updates
3. Bell icon shows red dot
↓
User taps bell/notification:
↓
IF notification type == "order_update"
    → SCREEN 13 (Order Tracking)
IF notification type == "advisory"
    → SCREEN 20 (Voice Player)
```

IF notification type == "offer"

→ SCREEN 8 (Products with filter)

SEARCH FLOW

text

From SCREEN 8 (Product List):

↓

TAP search bar → Keyboard opens

↓

Type search term → Real-time filtering

↓

TAP product → SCREEN 9

OR

TAP "Filter" icon → SCREEN 34 (Filters)

├── Price range

├── Category

└── Stock availability

🚨 ERROR & EDGE CASE FLOWS

NETWORK OFFLINE

text

TRY any network action:

↓

IF offline:

SHOW snackbar: "No internet"

For orders: "Queued for sync"

Products: Show cached version

Advisory: Show last loaded

↓

WHEN back online:

Auto-sync pending orders

Refresh product stock

LOW STOCK/OUT OF STOCK

text

User views SCREEN 9 (Product Details):

↓

IF stock == 0:

  SHOW "Out of Stock" badge

  Disable "Add to Cart"

  SHOW "Notify when available" button

↓

IF stock < 5:

  SHOW "Low Stock" warning

  Allow add to cart

CREDIT LIMIT EXCEEDED

text

Farmer selects "Add to Credit" payment:

↓

CHECK credit balance + order total

↓

IF exceeds limit:

  SHOW alert: "Credit limit reached"

  Disable "Place Order"

  Offer "Cash on Pickup" option

📊 DATA SYNC FLOW

REAL-TIME UPDATES

text

1. Product stock changes → Update all users viewing that product

2. Order status changes → Update farmer's order tracking

3. New advisory → Push to relevant farmers

4. Price changes → Update cart if not checked out yet

BACKGROUND SYNC

text

App in background:

```
├── Receive FCM notifications
├── Update local cache
└── Prepare data for next open
```

🔐 SECURITY & ACCESS FLOW

ROLE-BASED REDIRECTION

text

Farmer tries to access /owner/* → Redirect to SCREEN 6

Owner tries to access /farmer/* → Redirect to SCREEN 22

Unauthenticated tries any screen → Redirect to SCREEN 2

SESSION MANAGEMENT

text

Token expires → Auto logout → SCREEN 2

Force logout from owner → Farmer logged out

Change phone → Logout all devices

🎯 COMPLETE SCREEN LIST WITH IDs

text

AUTH FLOW:

SC1 - Splash

SC2 - Language Select

SC3 - Phone Login

SC4 - OTP Verify

SC5 - Auto Route (invisible)

FARMER FLOW:

SC6 - Farmer Home

SC7 - Categories

SC8 - Product List

SC9 - Product Details

SC10 - Cart

SC11 - Checkout

SC12 - Order Success

SC13 - Order Tracking

SC14 - Re-Buy

ADVISORY FLOW:

SC15 - Add Crop

SC16 - My Crops

SC17 - Recommended Products

SC18 - Dosage Calculator

SC19 - Advisory Messages

SC20 - Voice Player

SC21 - Pest Alerts

OWNER FLOW:

SC22 - Owner Dashboard

SC23 - Order Management

SC24 - Stock Management

SC25 - Add/Edit Product

SC26 - Farmer Accounts

SC27 - Credit Management

SC28 - Broadcast Alerts

SC29 - Reports

PROFILE & SETTINGS:

SC30 - Profile

SC31 - Crop Details

SC32 - Order Details (Owner)

SC33 - Farmer Details (Owner)

SC34 - Filters

SC35 - Notifications

SC36 - Edit Profile

🎛️ BOTTOM NAVIGATION STATES

FARMER NAVIGATION

text

HOME (SC6) - Active when on home/dashboard

PRODUCTS (SC8) - Active on product listings/details

ADVISORY (SC19) - Active on advisory/voice/crops

PROFILE (SC30) - Active on profile/settings

OWNER NAVIGATION

text

DASHBOARD (SC22) - Business overview

ORDERS (SC23) - Order processing

STOCK (SC24) - Inventory management

FARMERS (SC26) - Customer accounts

🚀 LAUNCH CONFIGURATIONS

DEEP LINKS SUPPORT

text

myfarmapp://product/{id} → SC9

myfarmapp://order/{id} → SC13

myfarmapp://advisory/{id} → SC20

myfarmapp://offer/{id} → SC8 with filter

PUSH NOTIFICATION PAYLOADS

text

```
{
  type: "order_update",
  orderId: "123",
  screen: "order_tracking"
```

}

✅ COMPLETE NAVIGATION MATRIX

FROM SCREEN	POSSIBLE DESTINATIONS
Splash	Language (if new) OR Home (if logged in)
Home	Products, My Crops, Re-Buy, Orders, Profile
Products	Product Details, Cart, Filters
Product Details	Cart, Dosage Calculator, Home
Cart	Checkout, Continue Shopping (Home)
Checkout	Order Success, Back to Cart
Order Success	View Orders OR Home
My Crops	Add Crop, Crop Details, Recommended Products
Advisory Messages	Voice Player, Product List
Owner Dashboard	All owner screens
Profile	All farmer screens, Logout